

| Education           |                        |                                                |         |
|---------------------|------------------------|------------------------------------------------|---------|
| Year                | Degree/Certificate     | Institute                                      | CPI/%   |
| Oct 2022 – Aug 2026 | B.Tech in CS & IT      | Ajay Kumar Garg Engineering College, Delhi NCR | 7.25/10 |
| 2021 – 2022         | Senior Secondary (XII) | New Era School, Delhi NCR                      | 91.2%   |

Experience

Software Engineer | SYNRA Inc. , Tokyo, Japan

(August 2026 – Present)

RAG-Backed Analysis Platform & Cross-Platform Capture Client

Python, LangChain, FastAPI, Flutter, Swift, C++

- Retrieval Pipeline:** Built a RAG layer over vibration reports with FastAPI and LangChain.
- Capture Client:** Shipped Flutter and iOS apps with locked exposure, focus and frame timing.

Google Summer of Code 2025 Contributor | P4 Language Consortium, Stanford, CA

(May 2025 – Sept 2025)

SpliDT Project: Scaling Stateful Decision Tree Algorithms in P4

P4, Python, P4Runtime, Docker, C++, Intel Tofino

- Switch-Native Architecture:** Compiled ML decision trees directly into P4 data planes with TCAM-efficient Subtree IDs to resolve hardware ASIC constraints.
- Validation & Optimization:** Built a P4Runtime testing pipeline (Tofino/Mininet) improving reliability by **30%**; hit **10 Gbps** line rate with **20%** lower inference latency.

Open Source Contributions

- Project-HAMi/HAMi:** Fixed scheduler and device-plugin bugs in HAMi, a CNCF project for GPU/device resource sharing and isolation across Kubernetes clusters - **5 PRs** merged.
- WasmEdge/WasmEdge:** [#4470](#) fix(wasi): resolve **100%** CPU usage on macOS due to clock mismatch; [#4466](#) feat(docker): add standalone alpine-base image build pipeline.
- facebook/bpfilter:** [#507](#) build: update clang-tidy brace config, fix opts.c/set.c, and NOLINT dump.c.

Key Projects

Cutable | Go, Next.js, PostgreSQL, OpenRouter, E2B, WebSockets, Nomad

(Project Link)

**Objective:** Generate editable React applications from text and visual requirements.

**Approach:** Orchestrated OpenRouter tools and E2B sandboxes in Go; streamed code and live previews to Next.js over WebSockets.

**Impact:** Ships multimodal BYOK generation, persistent projects, and health-gated deployments.

DMIE | Next.js, TypeScript, Python, FastAPI, Vector DB, LLM

(Project Link)

**Objective:** Determine whether a startup idea already exists by comparing it against more than **500,000** company records.

**Approach:** Built a scheduled scraper that keeps the startup dataset current; combined LLM query with vector search and backend filters.

**Impact:** Raised USD **10,000** and serves more than **100** users worldwide.

Moodish | JavaScript, Node.js, OpenAI API, Swiggy MCP

(Project Link)

**Objective:** Cross-platform food recommendation app with personal and group-order flows.

**Approach:** Flutter Android and iOS clients across **6** tagged releases, with Google OAuth deep-linking and a live group-order lobby; Deterministic scoring over craving, budget and dietary constraints with an; inspectable recommendation trace.

**Impact:** Produces safer, inspectable food recommendations using Swiggy MCP

Myprod | Go, Nomad, Traefik, WireGuard, Docker

(Project Link)

**Objective:** Turn manually provisioned VPS machines into a low-overhead personal compute pool for backend services.

**Approach:** Built a Go CLI and operator dashboard over Nomad scheduling, Traefik ingress, and WireGuard networking, with guarded node lifecycle controls and live resource metrics.

**Impact:** Provides one control surface for deploying services across heterogeneous free-tier and temporary compute.

Technical Skills

- Languages:** Python, Go, TypeScript, JavaScript, C++, Dart, Swift
- AI & Orchestration:** OpenRouter, Azure OpenAI, MCP, E2B, LLM tool-calling, vector search, Sklearn
- Backend & Data:** FastAPI, NodeJS, gRPC, Protobufs, WebSockets, PostgreSQL, Redis, MongoDB
- Tools / Platforms:** Docker, Nomad, CI/CD, Next.js, React, Linux, Kubernetes, eBPF, P4, Linux

Positions of Leadership

- Technical Writer, Cloud Native Community Group, New Delhi** (May 2024 – Dec 2024)
  - Led **6+** writers and **4** designers on documentation and event content; social media revamp lifted engagement **60%**.
- Co-Organizer & Anchor, Google Developer Groups & GDSC WOW 2024, Delhi NCR** (Nov 2023 – Nov 2025)
  - Co-led **10+** tech events; anchored GDSC WOW at IIIT Delhi (**800+** live audience, **31+** crew).

Publications

- R. Bharadwaj, **S. Jha**, V. Malyan, R. Gupta, “Trusture: A Blockchain-Based Decentralized Framework for Secure, Transparent, and Auditable NGO Transactions,” *2026 ICCSC*, Ghaziabad, India, Feb. **2026**. DOI: 10.1109/ICCSC67078.2026.11468597

Certifications & Honors

- 2x** AKTU State TT Gold (**2023, 2024**); Robotics Olympiad Bronze, Thailand (**2019**); NASA, NPTEL & ISRO certifications